

Assignment-2 Conditional statements

1. Task: Calculate Profit Percentage

- Write a program that takes input for the **cost price** and **selling price** of an item.
 - **Hints**
 - Prompt the user to input the **cost price** and **selling price**.
 - Determine whether the transaction resulted in a **profit** or **loss**.
 - If there is a profit calculate the **profit percentage**; if there is a loss calculate the **loss percentage**.
 - Display the profit or loss and the respective percentage.
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2. Task: Cricket Stats Analyzer

- **Objective:** Write a script to analyze cricket stats for a team.
 - **Hints:**
 - Prompt the user to input the runs scored by each of the five players in a cricket match.
 - For each player (Player 1 to Player 5) ask the user to input the runs they scored.
 - Calculate the **total runs** scored by all players and the **average runs**.
 - Display the **total runs** and **average runs** to the user.
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3. Task: Retirement Age Calculator

- **Objective:** Write a program that prompts the user for their **age** and tells them how many years until they reach retirement age (65).
 - **Hints:**
 - Ask the user to input their **age**.
 - Calculate how many more years they have until they reach **65 years** of age.
 - Display the number of years left until retirement or a message if the user has already reached retirement age.
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4. Task: Calculate the Area of a Circle

- **Objective:** Write a program to calculate the **area of a circle**.
 - **Hints:**
 - Ask the user to input the **radius** of the circle.
 - Calculate the **area** of the circle using the formula: $\text{Area} = \pi * \text{radius}^2$.
 - Display the calculated **area**.
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5. Task: Salary Calculation

- **Objective:** You have to calculate an employee's salary by computing the gross salary tax and net salary based on the given parameters.
- **Hints:**
 - **Base Salary** = ₹50000
 - **Bonus** = ₹5000
 - **Tax Rate** = 10%
 - **Other Charges** = ₹2000

Display the **Gross Salary Tax** and **Net Salary**.

6. Task: Bank Loan Approval System

- **Objective:** You have to create a javascript script that checks whether an user is eligible for a bank loan based on various criteria.
- **Hints:**
 - The applicant's **age** must be between 18 and 60 years.
 - The applicant's **monthly income** must be greater than or equal to ₹25000.
 - The applicant's **credit score** must be greater than or equal to 700.
 - The applicant must not have any **outstanding debts** greater than ₹10000

1. Output:

- Display "Loan Approved" if the applicant meets all the conditions.
- Otherwise display "Loan Rejected".

7. Task: Students Interview Eligibility Checker

- **Objective:** you have to design a javascript script that checks whether a student is eligible for an interview based on their academic score attendance percentage and extracurricular participation.
- **Input:**
 - **Academic Score (percentage):** A floating-point number representing the student's academic score. Ex .78.88
 - **Attendance Percentage:** A floating-point number representing the student's attendance percentage. Ex.85.88
 - **Extracurricular Participation:** This indicates whether the student has participated in any extracurricular activities. Ex.Yes/no

Conditions for Interview Eligibility:

1. The student's **academic score** must be **60 or above**.
2. The student's **attendance percentage** must be **75 or above**.
3. The student should have participated in **at least one extracurricular activity**.

Output:

- If the student meets all three conditions print "**Eligible for Interview**".
 - If the student fails to meet any of the conditions print "**Not Eligible for Interview**".
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8. Task: Validating Email Domain

- **Objective:** You will implement a javascript program to validate the domain of a user's email address. The program will check if the email contains a specific domain (e.g. "gmail.com").

Problem Statement:

You are building a registration system that only accepts email addresses from a certain domain (e.g. "gmail.com"). Your task is to:

1. Prompt the user to enter their email address.
2. Check if the entered email address contains the domain "**gmail.com**".
3. Display whether the email is eligible for registration based on the domain check.
4. Print a message to inform the user if their email is eligible for registration or not.

10.Task : Student Grading System

Create a javascript program to calculate a student's grade based on their marks.

Task:

1. **Input:** Prompt the user to enter their marks.
2. **Criteria:**
 - Grade **A**: 90–100
 - Grade **B**: 80–89
 - Grade **C**: 70–79
 - Grade **D**: 60–69
 - Grade **E**: 50–59
 - Grade **F**: 0–49
 - Invalid marks: Outside the range 0–100.
3. **Output:** Display the grade or an error message for invalid marks.

Example Outputs:

- Marks: 85 → Grade: **B**
- Marks: 45 → Grade: **F**
- Marks: 105 → **Invalid marks.**

11.Task : Authentication System.

Write a javascript program that authenticates a user by checking their username and password. The program should compare the entered credentials with predefined valid credentials.

- **Predefined valid usernames and corresponding passwords**

```
username1 = "user1"
```

```
username1_password1 = "pass@123"
```

Instructions:

1. **Input:**
 - Prompt the user to input their username and password.
 2. **Processing:**
 - Check if the username and password match
 3. **Output:**
 - If both the username and password match the predefined valid credentials display "Authentication successful."
 - If either the username or the password does not match display "Authentication failed."
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12. Employee Salary Based on Experience.

You are building a system for a **Human Resources (HR) department** that calculates an employee's salary based on their years of experience. The system assigns salary tiers based on the number of years an employee has been working. It also offers bonuses for employees who have more than 15 years of experience.

Scenario Breakdown:

Context 1: Senior Employee

- An employee with **10 or more years of experience** is classified as a **Senior Employee**. The base salary for such an employee is **80000**.
- If the employee has more than **15 years of experience** they receive a **bonus of 5000** to their salary.

Example:

An employee with **18 years of experience**:

- They are classified as Senior **Employees**.
- Their base salary is **80000**.
- Since they have more than **15 years of experience** they receive an additional **5000 bonus**.

- The final salary is **85000**.

Context 2: Mid-Level Employee

- Employees with **5 to 9 years of experience** are classified as **Mid-Level Employees**.
- Their base salary is **50000** with no bonus.

Example:

An employee with **7 years of experience**:

- They are classified as a **Mid-Level Employee**.
- Their base salary is **50000**.
- Since they have fewer than 10 years of experience no bonus is added.
- The final salary is **50000**.

Context 3: Junior Employee

- Employees with **less than 5 years of experience** are classified as **Junior Employees**.
- Their base salary is **30000** with no bonus.

Example:

An employee with **3 years of experience**:

- They are classified as Junior **Employees**.
- Their base salary is **30000**.
- No bonus is added.
- The final salary is **30000**.

Output Examples:

Senior Employee with 18 years of experience:

Enter years of experience: 18

Senior employee

Experience exceeds 15 years. Bonus added.

Salary: 85000

Mid-Level Employee with 7 years of experience:

Enter years of experience: 7

Mid-level employee

Salary: 50000

Junior Employee with 3 years of experience:

Enter years of experience: 3

Junior employee

Salary: 30000

13. Library Charge Calculation

Problem Statement:

Write a javascript program that calculates the library charge based on the number of days a book has been borrowed.

Charge Criteria:

- Up to 5 days: Rs. 2 per day
- 6 to 10 days: Rs. 3 per day
- 11 to 15 days: Rs. 4 per day
- More than 15 days: Rs. 5 per day

Instructions:

1. **Input:** Prompt the user to enter the number of days the book has been borrowed.
2. **Processing:** Calculate the charge based on the given criteria.
3. **Output:** Display the calculated charge.

14. Arranging Three Numbers in Descending Order

Task:

Write a javascript program to arrange three numbers in descending order.

Input:

Prompt the user to enter three numbers.

Processing:

Sort the numbers in descending order.

Example:

- Enter first number: 3
- Enter second number: 1
- Enter third number: 2

Output:

- Numbers in Descending Order: 3, 2, 1
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15. Tax Calculation for Car Purchase

Write a program to calculate the **tax** on a car purchase based on the car brand and its price.

1. **Mahindra**: 5% tax for prices between 7L (7 lakh) and 10L.
 2. **Audi**: 10% tax for prices between 10L and 15L.
 3. **Jaguar**: 25% tax for prices between 15L and 20L.
 4. **Mercedes**: 30% tax for prices between 20L and 25L.
 5. Input: The car brand and price.
 6. Output: The calculated tax on the purchase.
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16. Finding the Middle Number

- **Task**: Write a program to determine the middle number among three inputs.
 - **Input**: Prompt the user to enter three numbers.
 - **Processing**: Identify the middle number, which is neither the largest nor the smallest.
 - **Output**: Display the middle number.
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17. Find the greatest number.

- **Task:** Write a program to find greatest number from three number
 - **Input:** Prompt the user to enter three numbers.
 - **Output:** Display the greatest number.
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18.Authentication System

- **Task:** Write a program to authenticate a user by validating their username and password.
 - **Predefined Credentials:**
 - Username: `user1`
 - Password: `pass@123`
 - **Input:** Prompt the user to input their username and password.
 - **Output:**
 - If the credentials match, display "Authentication successful."
 - If they do not match, display "Authentication failed."
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19.Calculate Class Attendance Percentage

- **Task:** Write a program to calculate the percentage of classes attended by a student and determine their eligibility to sit in the exam.
 - **Conditions:**
 - Attendance percentage < 75%: Not eligible to sit in the exam.
 - Attendance percentage ≥ 75%: Eligible to sit in the exam.
 - **Output:** Display the attendance percentage and eligibility status.
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20.Library Charge Calculation

- **Task:** Write a program to calculate the library charges based on the number of days a book has been borrowed.
- **Charge Criteria:**
 - Up to 5 days: ₹2/day.
 - 6 to 10 days: ₹3/day.
 - 11 to 15 days: ₹4/day.
 - More than 15 days: ₹5/day.
- **Output:** Display the total charges.

21. UPSC Selection Process

- **Task:** Simulate the UPSC selection process with the following steps:
 1. **Eligibility Check**
 - **Criteria:**
 - Age: 21–32 years.
 - Graduate status: Must be a graduate.
 - Nationality: Must be "Indian".
 - **Output:**
 - If eligible, proceed to Prelims.
 - If ineligible, display the reason for ineligibility.
 2. **Prelims Exam**
 - **Processing:** Check if the candidate's score $\geq$ cut-off.
 - **Output:**
 - If passed, proceed to Mains.
 - If failed, display "You failed the Prelims."
 3. **Mains Exam**
 - **Processing:** Check if the candidate's score $\geq$ cut-off.
 - **Output:**
 - If passed, proceed to Interview.
 - If failed, display "You failed the Mains."
 4. **Interview**
 - **Processing:** Check if the candidate's score $\geq$ cut-off.
 - **Output:**
 - If passed, display "Congratulations! You have cleared the UPSC."
 - If failed, display "You failed the Interview."
- **Final Output:** Use nested conditional statements to simulate the entire process.

22. Menu-Driven Login System

1. Create the Menu:

- Display a menu with three choices for the user:
 - Login with Phone
 - Login with Email

- Exit the system

2. Predefined Credentials:

- Phone number: "1234567890"
- OTP: "1234"
- Email: "user@example.com"
- Password: "password123"

3. Login Functionality:

- **Option 1 (Login with Phone):**
 - Prompt the user to enter their phone number and OTP.
 - Compare the input with a predefined phone number and OTP.
 - Display success if both match or an error message if they don't.
- **Option 2 (Login with Email):**
 - Prompt the user to enter their email and password.
 - Compare the input with predefined email and password.
 - Display success if both match or an error message if they don't.
- **Option 3 (Exit):**
 - Display an exit message and terminate the program.
- **Invalid Input:**
 - Handle invalid user choices and ask the user to select a valid option.

Output:

1. If the user enters a valid phone number and OTP, display: "Login successful with phone!"
2. If the user enters valid email and password, display: "Login successful with email!"
3. If the user selects the exit option, display: "Exiting the program. Have a nice day!"
4. If the user enters invalid credentials or an invalid choice, display appropriate error messages.

23 .Create Your Own KBC Game

Design and implement a quiz game inspired by the popular *Kaun Banega Crorepati (KBC)* game show. The aim of this assignment is to test the user's knowledge through a series of multiple-choice questions, track their score, and display statistics at the end of the game. The game also provides the flexibility to skip any question.

Instructions:

1. Game Structure:

- The game will consist of **5 multiple-choice questions**.
- The user will be asked a question with **4 options** (A, B, C, D).
- The user can choose to **skip** any question they do not want to answer.

2. Scoring System:

- Points will be awarded for **correct answers** as follows:
 - **Question 1** → 1000 points
 - **Question 2** → 2000 points
 - **Question 3** → 3000 points
 - **Question 4** → 5000 points
 - **Question 5** → 10000 points
- For **incorrect answers**, no points will be awarded.
- For **skipped questions**, no points will be awarded, but the game will continue.

3. End of Game Statistics:

- At the end of the game, the following **statistics** will be displayed:
 - **Total score** accumulated from correct answers.
 - **Number of correct answers** provided by the user.
 - **Number of skipped questions**.
 - **Number of wrong answers**

4. User Experience:

- At the beginning of the game, ask the user whether they would like to **start or not** the game.
- Provide the option for the user to **skip** any question at any point..

5. Game Ending:

- The game will end when all the questions have been answered or skipped. The user should receive their total score and a summary of their performance.